CMAI 480

Computer Vision

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Reading Assignment

- Szeliski, Chapter 1 (Introduction)

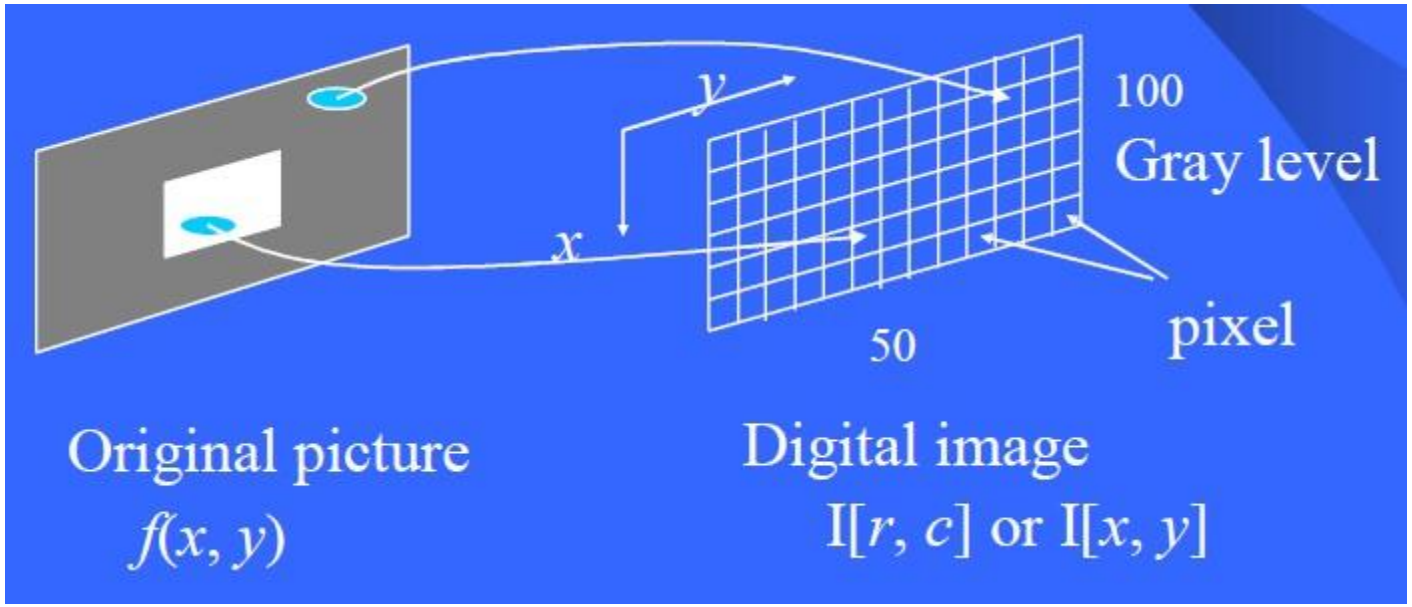
Introduction about Computer Vision and its relationship to other field

Computer Vision, What is it?

- Computer Vision: the study of analysis of image/video content in order to achieve results similar to those as by people.

What Is a Digital Image?

- Is composed of a finite number of elements each of which has a particular location and value (pixels, picture elements).



An Example of a Digital Image



(a) Color representation



(b) Red component



(a) Monochrome representation



(c) Green component



(d) Blue component



(d) *RGB* sum edge map

Digital Image Processing: PIKS Inside, Third Edition. William K. Pratt Copyright © 2001 John Wiley & Sons, Inc.

Several fields deal with images

INPUT		OUTPUT	
		IMAGE	**DESCRIPTION**
	IMAGE	Image Processing	Computer Vision
	DESCRIPTION	Computer Graphics	Data Mining, ML, or AI.

Image Processing: What Is It?

- Image Processing: the enhancement or other manipulation of the image – the result of which is usually another image.

- Processing images by means of a digital computer.
  - Image acquisition.
  - Representing, storing and displaying images.
  - Image transformations
  - Image filtering, enhancement and restoration
  - Image compression
  - Image segmentation

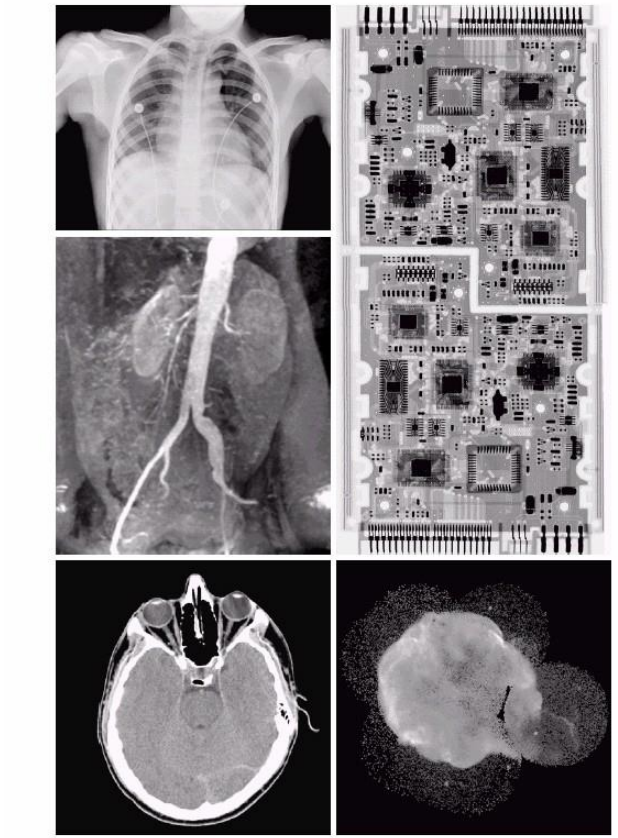
Image Processing: What Is It?

- Image processing is a subclass of signal processing concerned specifically with pictures.
- Improve image quality for human perception and/or computer interpretation.



Digital Image Processing (DIP)

“A picture is worth a thousand words”.

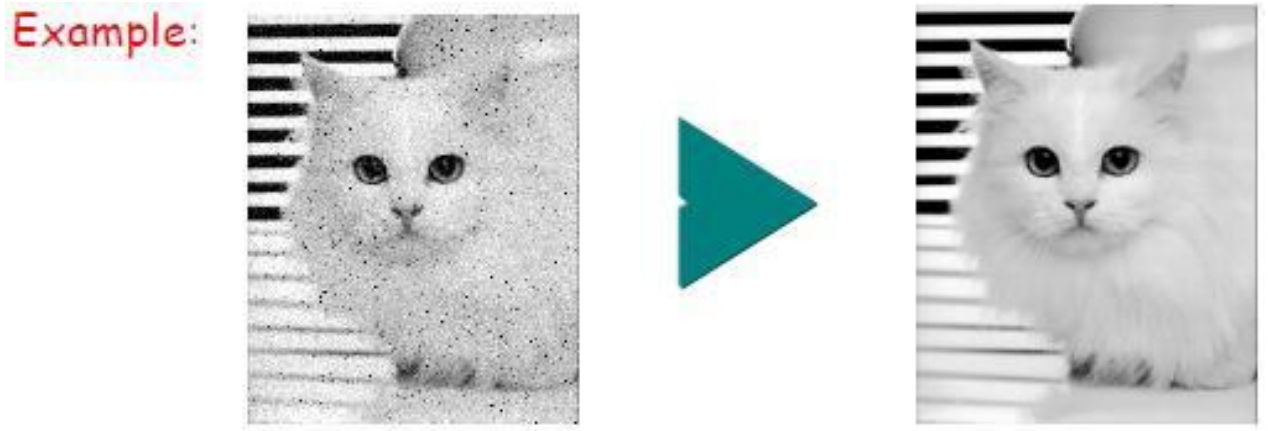


a b c d e

FIGURE 1.7 Examples of X-ray imaging. (a) Chest X-ray. (b) Aortic angiogram. (c) Head CT. (d) Circuit boards. (e) Cygnus Loop. (Images courtesy of (a) and (c) Dr. David R. Pickens, Dept. of Radiology & Radiological Sciences, Vanderbilt University Medical Center, (b) Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School, (d) Mr. Joseph E. Pascente, Lixi, Inc., and (e) NASA.)

Example DIP:Image Enhancement

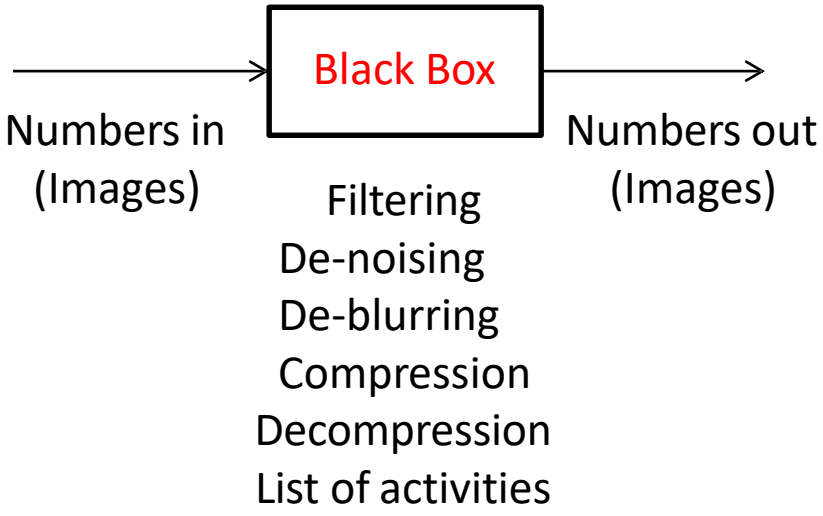
- To bring out detail is obscured, or simply to highlight certain features of interest in an image.



What is Computer Vision

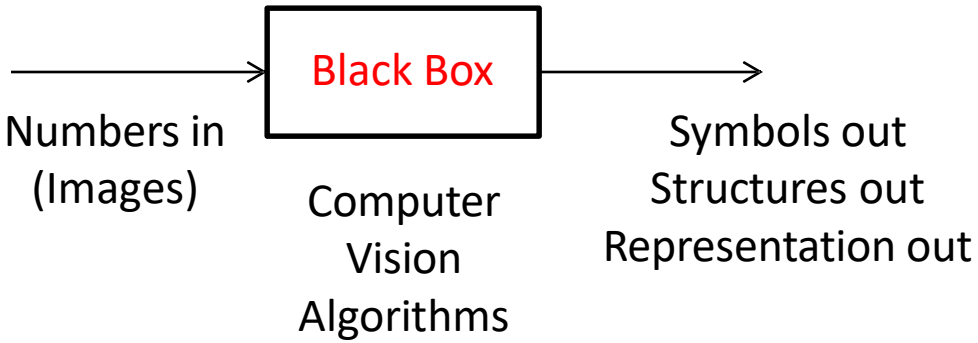
- Computer Vision and Digital Image Processing
  - Split Recently 20 ~ 25 years ago

- Digital Image Processing (DIP)



What is Computer Vision

- Computer Vision (CV)



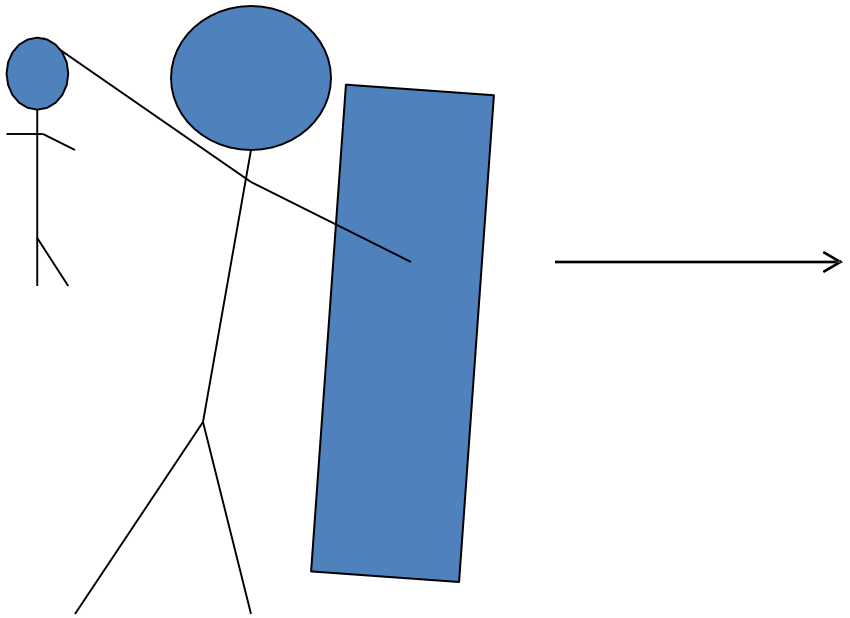
- What you will do with those representations
  - What algorithm to use to deal with those representation
  - That's what make CV open world ended (More Fun)

Computer Graphics: What Is It?

- Computer Graphics : the creation of images.

Computer Vision and Nearby Fields

- Computer Graphics: Models to Images





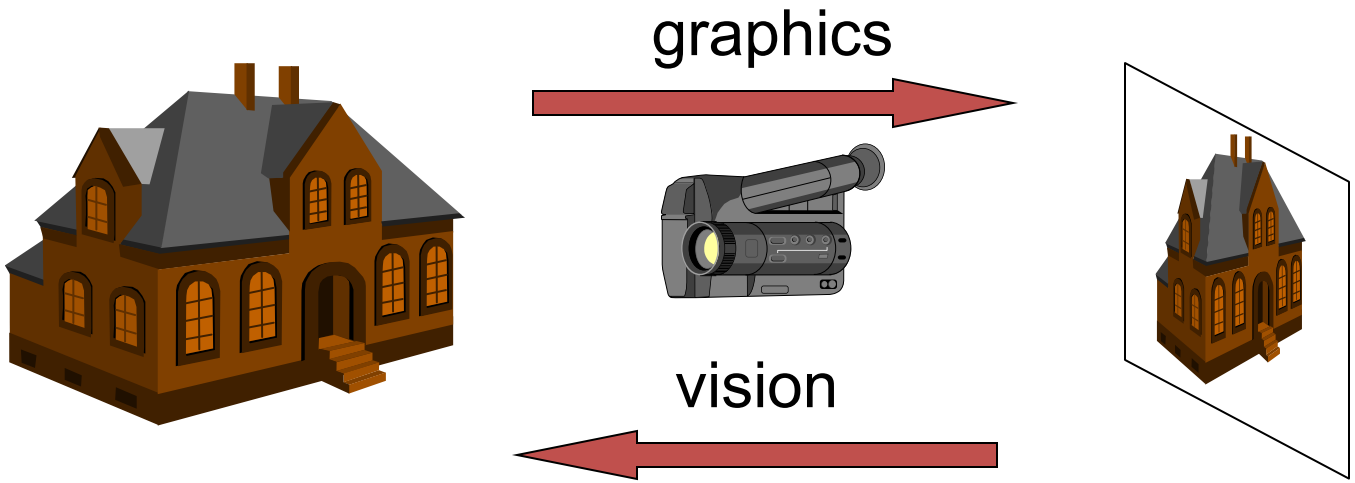
© Pixar
http://pichost.me/1578472/

Computer Vision vs. Graphics

3D→2D implies information loss

sensitivity to errors

need for models



Machine Learning: What Is It?

- **Machine learning** is a scientific discipline that is concerned with the design and development of algorithms that allow computers to change behavior based on data, such as from sensor data or databases (from Wikipedia)
- A major focus of machine learning research is to automatically learn to recognize complex patterns and make intelligent decisions based on data.

Computer Vision vs. Machine Learning

- Machine Learning is very useful for Computer Vision (e.g., learning for vision)
- Computer Vision is more than just learning
  - Modeling
  - Example based learning
- In Machine Learning, it usually does not care about how to obtain the data or sensors
- In Computer Vision, we care how to obtain the visual data (sensor design, active vision), how to represent the visual data, and others

Computer Vision: Images to models

Make computers “understand” images and video.



What kind of scene?

Where are the cars?

How far is the building?

…

Vision is really hard

- Vision is an amazing feat of natural intelligence
  - Visual cortex occupies about 50% of Macaque brain
  - More human brain devoted to vision than anything else



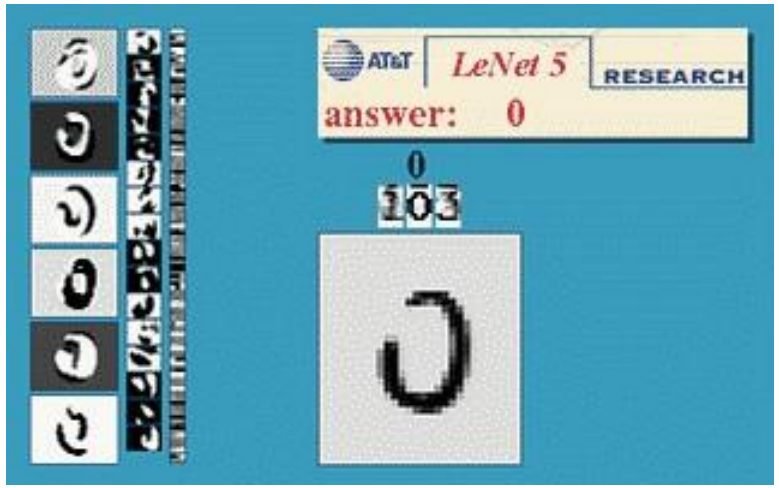
What is Computer Vision?
Examples of state-of-the-art

- What are examples of computer vision being used in the world?

Optical character recognition (OCR)

Technology to convert scanned docs to text

- If you have a scanner, it probably came with OCR software



Digit recognition, AT&T labs
http://www.research.att.com/~yann/



License plate readers
http://en.wikipedia.org/wiki/Automatic_number_plate_recognition

Face detection

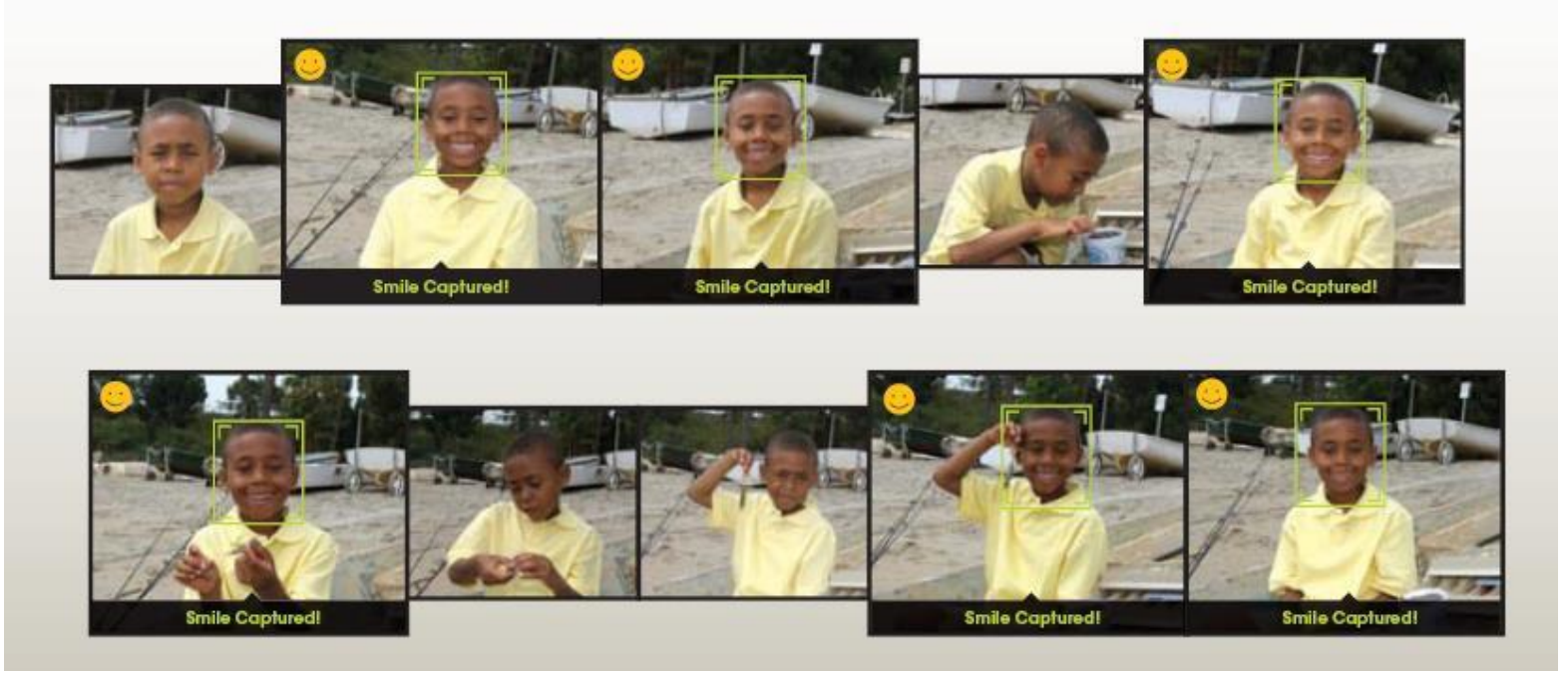


- Many new digital cameras now detect faces
  - Canon, Sony, Fuji, …

Smile detection

The Smile Shutter flow

Imagine a camera smart enough to catch every smile! In Smile Shutter Mode, your Cyber-shot® camera can automatically trip the shutter at just the right instant to catch the perfect expression.



[Sony Cyber-shot® T70 Digital Still Camera]

3D from thousands of images



Building Rome in a Day: Agarwal et al. 2009

Login without a password…





Fingerprint scanners on many new laptops, other devices





Face recognition systems now beginning to appear more widely
http://www.sensiblevision.com/

Object recognition (in mobile phones)



[Point & Find](), [Nokia]()

[Google Goggles]()

Special effects: motion capture



Pirates of the Carribean, Industrial Light and Magic

Sports



Sportvision first down line
Nice explanation on www.howstuffworks.com

http://www.sportvision.com/video.html

Smart cars



- Mobileye
  - Vision systems currently in high-end BMW, GM, Volvo models

Google cars



Oct 9, 2010. "Google Cars Drive Themselves, in Traffic". *The New York Times*. John Markoff
June 24, 2011. "Nevada state law paves the way for driverless cars". *Financial Post*. Christine Dobby
Aug 9, 2011, "Human error blamed after Google's driverless car sparks five-vehicle crash". *The Star* (Toronto)

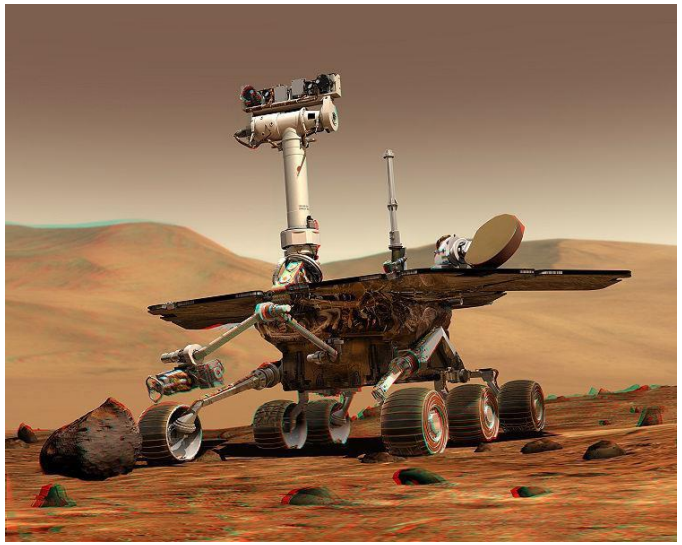
Industrial robots





Vision-guided robots position nut runners on wheels

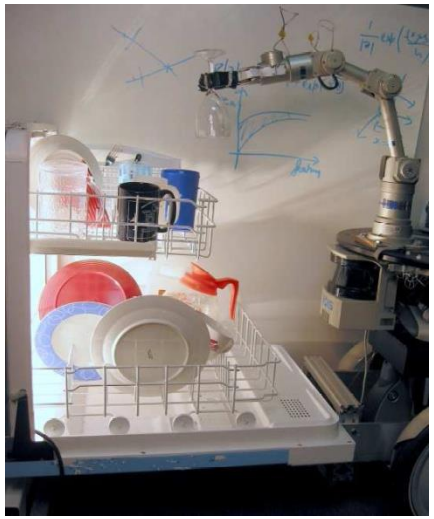
Mobile robots



NASA's Mars Spirit Rover
http://en.wikipedia.org/wiki/Spirit_rover

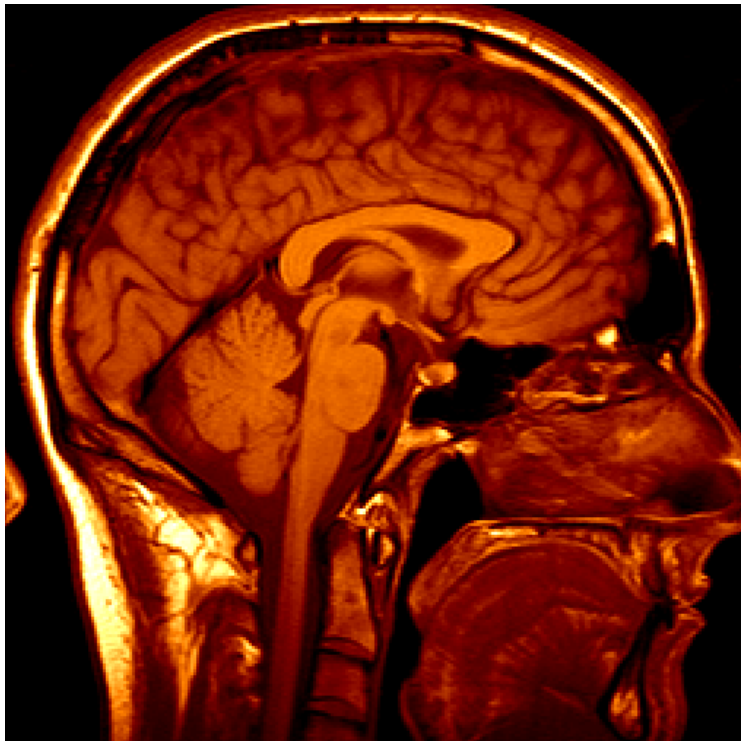


http://www.robocup.org/



Saxena et al. 2008
at Stanford

Medical imaging



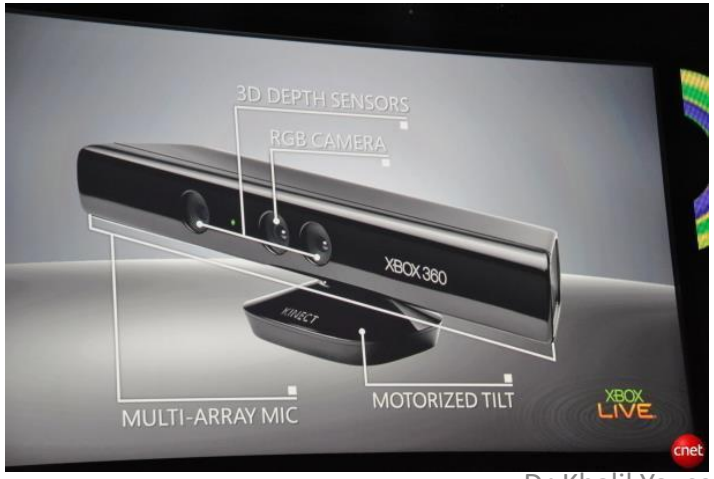
3D imaging
MRI, CT



Image guided surgery
Grimson et al., MIT

Interactive Games: Kinect

- Object Recognition: http://www.youtube.com/watch?feature=iv&v=fQ59dXOo63o
- Mario: http://www.youtube.com/watch?v=8CTJL5lUjHg
- 3D: http://www.youtube.com/watch?v=7QrnwoO1-8A
- Robot: http://www.youtube.com/watch?v=w8BmgtMKFbY





http://cs.brown.edu/courses/cs143/

WHY IS COMPUTER VISION DIFFICULT?



Viewpoint variation



Illumination



Scale

WHY IS COMPUTER VISION DIFFICULT?



Intra-class variation



Motion (Source: S. Lazebnik)



Background clutter

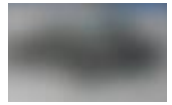


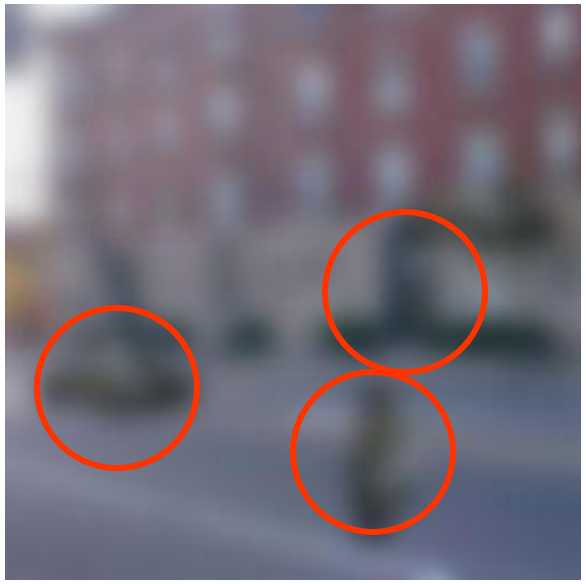
Occlusion

Challenges: local ambiguity







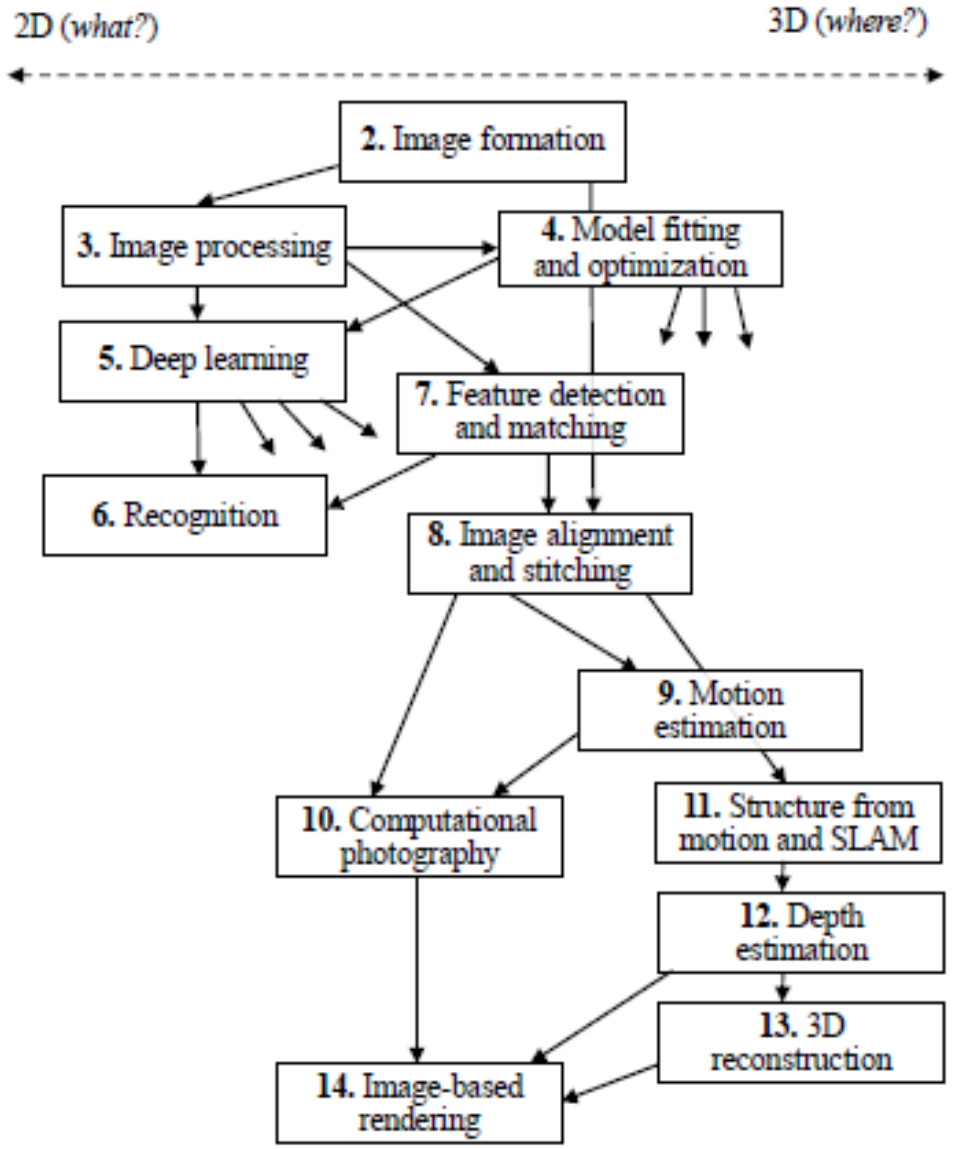




But there are lots of visual cues we can use…



The Course


2D (what?)
3D (where?)
2. Image formation
3. Image processing
4. Model fitting and optimization
5. Deep learning
7. Feature detection and matching
6. Recognition
8. Image alignment and stitching
9. Motion estimation
10. Computational photography
11. Structure from motion and SLAM
12. Depth estimation
13. 3D reconstruction
14. Image-based rendering

Next

- Reading Assignment: Read Chapter 2
  - Image Formation

The end